Hematology

ISSN: 1607-8454 (Online) Journal homepage: www.tandfonline.com/journals/yhem20

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To cite this article: Harinder Gill, Anskar Y. H. Leung, Chi-Chung Chan, June S. M. Lau, Candia Chan, Sze-Fai Yip, Herman Liu, Bonnie Kho, Vivien Mak, Harold K. K. Lee, Shek-Ying Lin, Chi-Kuen Lau & Yok-Lam Kwong (2016) Clinicopathologic features and prognostic indicators in Chinese patients with myelofibrosis, Hematology, 21:1, 10-18, DOI: 10.1179/1607845415Y.0000000045

To link to this article: https://doi.org/10.1179/1607845415Y.0000000045

Published online: 20 Aug 2015.

Clinicopathologic features and prognostic indicators in Chinese patients with myelofibrosis

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Objective: To define the clinicopathologic features, outcome, and prognostic indicators of myelofibrosis (MF) in Asian patients.
Methods: Two hundred and seventy consecutive Chinese patients (primary MF, $n = 207$; post-polycythemia vera MF, $n = 27$; and post-essential thrombocythemia MF, $n = 36$) from seven regional referral hospitals were analyzed.
Results: The median overall survival (OS) for primary MF was 66 months. Multivariate analysis showed that age >65 years ($P = 0.02$), platelet count $<100 \times 10^9$/l ($P = 0.001$), and leukemic transformation ($P = 0.001$) negatively impacted on OS. The median OS of 63 patients with secondary MF was 44 months. In primary MF, the 10-year cumulative risk of leukemic transformation was 28%. On multivariate analysis, unfavorable karyotypes significantly predicted inferior leukemia-free survival (LFS) ($P = 0.03$). In secondary MF, the 10-year cumulative risk of leukemic transformation was 31%. Circulating blasts ≥1% significantly predicted inferior LFS ($P = 0.04$). The international prognostic scoring system (IPSS) and dynamic IPSS were not significant survival predictors in our cohort. Eighteen patients underwent allogeneic hematopoietic stem cell transplantation. The median OS post-transplantation was merely 19 months.
Discussion: Platelet count $<100 \times 10^9$/l, unfavorable karyotypes, and circulating blasts >1% were negative prognostic indicators.
Conclusion: Chinese MF patients were similar to Western patients in clinicopathologic features and outcome.

Keywords: Myelofibrosis, Polycythemia vera, Essential thrombocythemia, Chinese

Introduction

Myelofibrosis (MF) is a myeloproliferative neoplasm (MPN), comprising primary MF and secondary MF evolving from polycythemia vera (post-PV MF) or essential thrombocythemia (post-ET MF).[1–3] MF has the worst prognosis among the Philadelphia-chromosome negative MPNs. The International Working Group for Myelofibrosis Research and Treatment (IWG-MRT) reported, in 1054 patients with primary MF, that the median survival was 69 months, with 31% of death attributed to secondary acute myeloid leukemia (AML).[4] Therefore, it is important to identify subgroup of patients with inferior survivals and high risk of leukemia transformation, both at diagnosis and early during the course of the disease. This will facilitate the selection of patients who require closer monitoring and referral for hematopoietic stem cell transplantation (HSCT).

Several prognostic models for survivals have been developed to assist treatment decisions at diagnosis and during the course of MF.[4–6] Interestingly, a recent report suggested possible differences in clinical manifestations and prognostic risk distribution among Chinese patients with primary MF, and a revised prognostic model for Chinese patients was proposed.[7] While these prognostic models predict survivals, there is a paucity of accurate indicators for AML transformation.

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DOI 10.1179/1607845415Y.0000000045

In this study, we reported a long-term follow-up study on 270 patients with primary MF, post-PV MF, and post-ET MF. Clinicopathologic features, natural history, treatment outcome, and prognostic factors for survivals were identified. The validity of the proposed prognostic model for Chinese patients was also examined in our primary MF cohort.

Materials and methods

Patients

Consecutive patients with primary MF, post-PV MF, and post-ET MF diagnosed between January 1990 and December 2013 were identified from seven regional hospitals, which served the entire population of Hong Kong. The diagnoses were reviewed based on the World Health Organization 2008 classification and the IWG-MRT criteria.[1–3] Data on clinicopathologic features, karyotype, *JAK2* V617F mutation, acquisition of transfusion dependence, treatment characteristics, and outcome were collected. The presence of hepatomegaly and splenomegaly was based on physical examination at diagnosis. The international prognostic scoring system (IPSS) and dynamic IPSS (DIPSS) were determined as previously described.[4,5] The IPSS and DIPSS had been proposed to be modified for Chinese patients[7] and were also determined in our patients. The modified IPSS and DIPSS for Chinese patients score the low-, intermediate-1, intermediate-2, and high-IPSS/DIPSS groups, respectively, as 0, 1, 2, and 3, and add a score of 1 each for no palpable splenomegaly and platelet count $\leq 100 \times 10^9$/l, resulting in three prognostic groups (scores for low: 0–1; intermediate: 2–3; and high: 4–5).[7] Unfavorable karyotypes included +8, −7/7q−, i(17q), inv(3), −5/5q−, 12p−, 11q23 rearrangements, and complex aberrations.[6,8] Treatment was decided at the discretion of the attending physician. We instituted red blood cell (RBC) transfusion with hemoglobin (Hb) level of <8 g/dl. Transfusion dependence was defined as the need of at least one unit of RBC transfusion once every 4 weeks to correct anemia for a duration of longer than 6 months.

Treatment

Treatment aimed at controlling leucocyte and platelet counts in primary MF. In post-PV and post-ET MF, treatment could be for reduction of Hb or platelet count. The most common initial treatment was hydroxyurea, followed by anagrelide, busulfan, melphalan, thalidomide, and radioactive phosphorus.

Definitions and statistical analyses

All data were censored on 1 October 2014. Overall survival (OS) was defined as the time from the diagnosis of primary MF, post-PV MF, and post-ET MF to death (event) or latest follow-up (censored). Blastic transformation or secondary AML was defined as blast ≥20% at any time during the disease course. Leukemia-free survival (LFS) was defined as the time from the diagnosis of primary MF, post-PV MF, and post-ET MF to secondary AML (event), death (censored), or latest follow-up (censored). Patients who underwent HSCT were censored at the time of transplantation. Comparison of categorical variables was performed using the Chi-square test. Continuous variables were compared with the Mann–Whitney *U* test. Prognostic factors for AML transformation were determined using logistic regression. Survival analyses and comparisons of survival were performed with the Kaplan–Meier method and log-rank test, respectively. Prognostic factors for survivals were determined using the Cox-proportional hazard regression model. Multivariate analyses were performed using logistic regression and Cox-proportional hazard regression for variables with *P*-values of <0.1 on univariate analysis. All data were analyzed using the SPSS 18.0 software (IBM, NY, USA), with two-tailed *P*-values of <0.05 regarded as statistically significant.

Results

Clinicopathologic features

Two hundred and seventy patients (primary MF, $n = 207$; post-PV MF, $n = 27$; and post-ET MF, $n = 36$), with a median duration of follow-up of 45 months (range: 1–247 months), were identified (Table 1). The *JAK2* V617 mutation study was performed in 93 patients. Karyotypes were available in 76 patients (supplementary file 1). Patients with primary MF were significantly different from those with secondary MF with respect to age >65 years ($P = 0.01$), median Hb ($P = 0.003$), Hb < 10 g/dl ($P = 0.001$), median circulating blast percentage ($P = 0.01$), circulating blasts ≥1% ($P = 0.002$), palpable hepatomegaly ($P = 0.002$), palpable splenomegaly ($P = 0.04$), *JAK2* V617F status ($P = 0.03$), prior use of cytoreduction ($P < 0.001$), and the median follow-up ($P = 0.008$) (Table 1).

Treatment

Hydroxyurea was used alone in 164 patients (primary MF, $n = 106$; post-PV MF, $n = 24$; and post-ET MF, $n = 34$). Other agents used together with hydroxyurea included anagrelide (primary MF, $n = 2$; post-PV MF, $n = 1$; and post-ET MF, $n = 3$), radioactive phosphorus (primary MF, $n = 1$; post-PV MF, $n = 1$), busulfan (primary MF, $n = 1$; post-PV MF, $n = 1$), oral melphalan (post-PV MF, $n = 1$; post-ET MF, $n = 1$), and thalidomide (primary MF, $n = 10$; post-ET MF, $n = 1$).

Outcome

The median OS of the entire cohort was 65 months (95% confidence interval, CI: 56–74). The 5-year and

Table 1 Clinicopathologic characteristics of 270 patients with MF

Clinicopathologic parameters	Primary MF	Post-PV MF	Post-ET MF	*P*-value
Total number	207	27	36	
Gender				
Male	123	14	20	
Female	84	13	16	0.71
Median age (range), years	68 (25–94)	64 (36–83)	60 (35–89)	0.15
Age >65 years	119	13	11	0.01
Presence of constitutional symptoms	66	10	15	0.48
Presentation blood counts				
Hb (median and range), g/dl	9.5 (2–17.5)	10.6 (6–19.5)	8.5 (5–11.4)	0.003
Hb <10 g/dl	116	11	30	0.001
White cell count (median, range), $\times 10^9$/l	9.6 (1.4–177.4)	15.6 (2.9–95.9)	9.5 (1.7–59.5)	0.07
White cell count $>25\times10^9$/l	25	3	8	0.24
Circulating blasts (median, range)	0 (0–15)	0 (0–5)	1 (0–8)	0.01
Circulating blasts >1%	53	11	19	0.002
Platelet count (median, range), $\times 10^9$/l	282 (7–2009)	269 (35–1108)	349 (6–877)	0.74
Platelet count $<100\times10^9$/l	51	2	6	0.09
IPSS				
Low	28	3	2	
Intermediate-1	57	8	8	
Intermediate-2	65	9	10	
High	57	7	16	0.51
DIPSS				
Low	28	3	2	
Intermediate-1	72	12	10	
Intermediate-2	88	11	18	
High	19	1	6	0.40
DIPSS-plus*				
Low	3	1	0	
Intermediate-1	13	0	2	
Intermediate-2	18	6	6	
High	20	2	4	0.37
Modified IPSS for Chinese				
Low	67	9	9	
Intermediate	106	17	21	
High	34	1	6	0.41
Modified DIPSS for Chinese				
Low	80	13	10	
Intermediate	109	4	24	
High	18	0	2	0.24
Red cell transfusion dependence	131	17	28	0.25
Palpable hepatomegaly	84	20	21	0.002
Palpable splenomegaly	160	26	31	0.04
Cytogenetic features				
Normal karyotype	31	4	8	0.60
Abnormal karyotype	24	5	4	0.60
Unfavorable karyotype	7	1	2	0.91
JAK2 V617F status†				
Present	35	11	6	
Absent	28	2	11	0.03

*Conventional cytogenetic analysis was performed in 76 patients (primary MF: 55; secondary MF: 21).
†*JAK2* V617F analysis was performed in 93 patients (primary MF: 63; secondary MF: 30).

10-year OS was 53 and 27%, respectively (Fig. 1A). The median LFS of the entire cohort was 240 months. The 5-year and 10-year LFS was 82 and 70%, respectively (Fig. 1B). The 5-year and 10-year cumulative risk of AML transformation was 18 and 30%, respectively (Fig. 1C). There were no statistically significant differences in OS and LFS between primary MF, post-PV MF, and post-ET MF ($P = 0.15$ and $P = 0.11$, respectively).

Modified IPSS and DIPSS for Chinese in primary MF

The outcome of the primary MF cohort stratified by the IPSS and DIPSS was shown in Fig. 2A and B. The low- and intermediate-1 risk groups, and the intermediate-2 and high-risk groups, overlapped and were not clearly separated. However, with the modified IPSS and DIPSS for Chinese, there was a clear separation of the low-, intermediate-, and high-risk groups (Fig. 2C and D).

Prognostic factors for OS in primary MF

The median OS was 66 (95% CI: 54–78) months (Fig. 3A). On univariate analysis (supplemental file 2), factors negatively impacting on OS included age >65 years ($P < 0.001$), Hb <10 g/dl ($P = 0.001$), platelet count $<100\times10^9$/l ($P < 0.001$), indetermediate-2 or high-risk IPSS ($P < 0.001$), intermediate-2 or

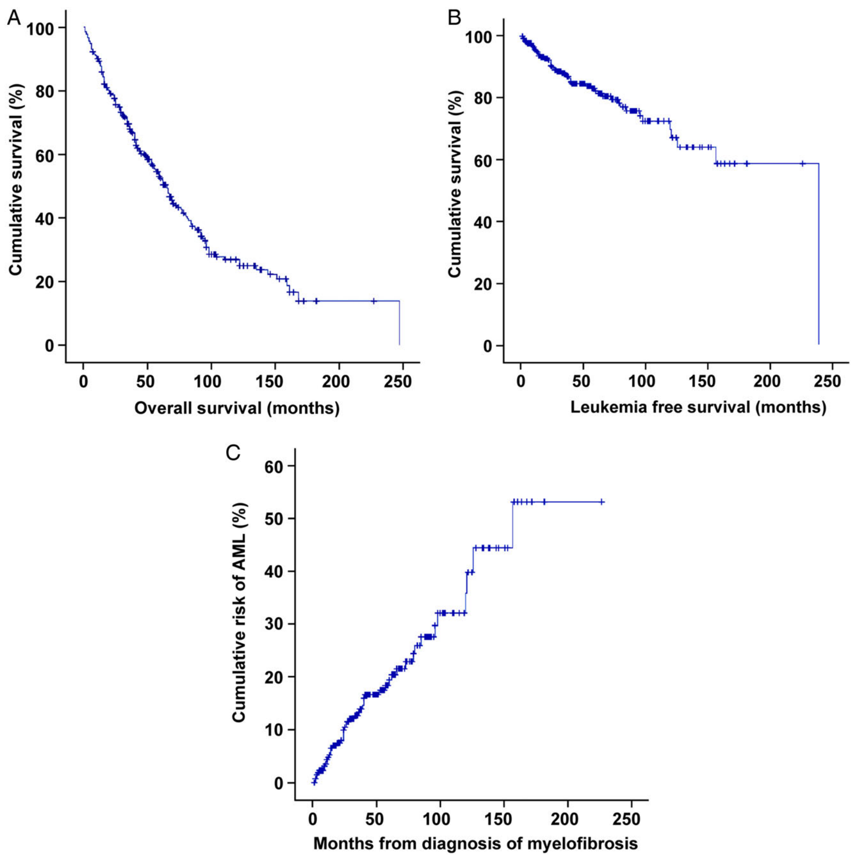

Figure 1 Outcome of 270 patients with MF. (A) OS. (B) LFS. (C) Cumulative risk of AML.

high-risk DIPSS ($P < 0.001$), transfusion dependence ($P < 0.001$), palpable hepatomegaly at diagnosis ($P = 0.017$), unfavorable karyotypes ($P = 0.03$), lack of prior cytoreduction ($P = 0.04$), and the development of secondary AML ($P < 0.001$) (supplemental file 3). On multivariate analysis, age >65 years ($P = 0.02$), platelet count $<100 \times 10^9/l$, and the development of secondary AML ($P = 0.001$) were associated with an inferior OS (Fig. 3B–D).

Prognostic factors for LFS in primary MF

The median LFS was not reached. The 5-year and 10-year LFS was 85 and 72%, respectively (Fig. 3E). The 5-year and 10-year cumulative risks of AML transformation were 15 and 28%, respectively (supplemental file 4). On univariate analysis (supplemental file 5), factors negatively impacting on LFS included circulating blasts ≥1% ($P = 0.005$), platelet count $<100 \times 10^9/l$ ($P = 0.002$), transfusion dependence ($P = 0.001$), palpable hepatomegaly at diagnosis ($P = 0.004$), and unfavorable karyotypes (0.002) (supplemental file 4). On multivariate analysis, unfavorable karyotypes remained the only negative prognostic factor ($P = 0.03$) (Fig. 3F).

Prognostic factors for OS in secondary MF

The median OS was 44 (95% CI: 30–58) months (Fig. 4A). On univariate analysis (supplemental file 6), platelet $<100 \times 10^9/l$ negatively impacted on OS ($P = 0.006$) (Fig. 4B). Because of the relatively small number of cases, multivariate analysis was not performed.

Prognostic factors for LFS in secondary MF

The median LFS was 240 months. The 10-year LFS was 69% (Fig. 4C). The 10-year cumulative risk of AML transformation was 31%. On univariate analysis (supplemental file 7), circulating blasts ≥1% ($P = 0.04$) were associated with an inferior LFS (Fig. 4D). Because of the small number of cases, multivariate analysis was not performed.

Outcome following allogeneic HSCT

Eighteen patients underwent allogeneic HSCT, with 15 patients achieving a complete remission. At a median follow-up post-HSCT of 14 months (range: 3–137 months), the median OS was 19 months. The 5-year OS post-HSCT was 49% (Fig. 5C).

Discussion

Data from MF have mainly been reported from Western populations, with very few studies conducted in Asia. This study analyzed a fairly large cohort of MF in an Asian population. There were only previously two other sizeable reports from Japan and China. In a Japanese study involving 202 patients, multivariate analysis showed that leucocyte count $>30 \times 10^9/l$ and the presence of cytogenetic

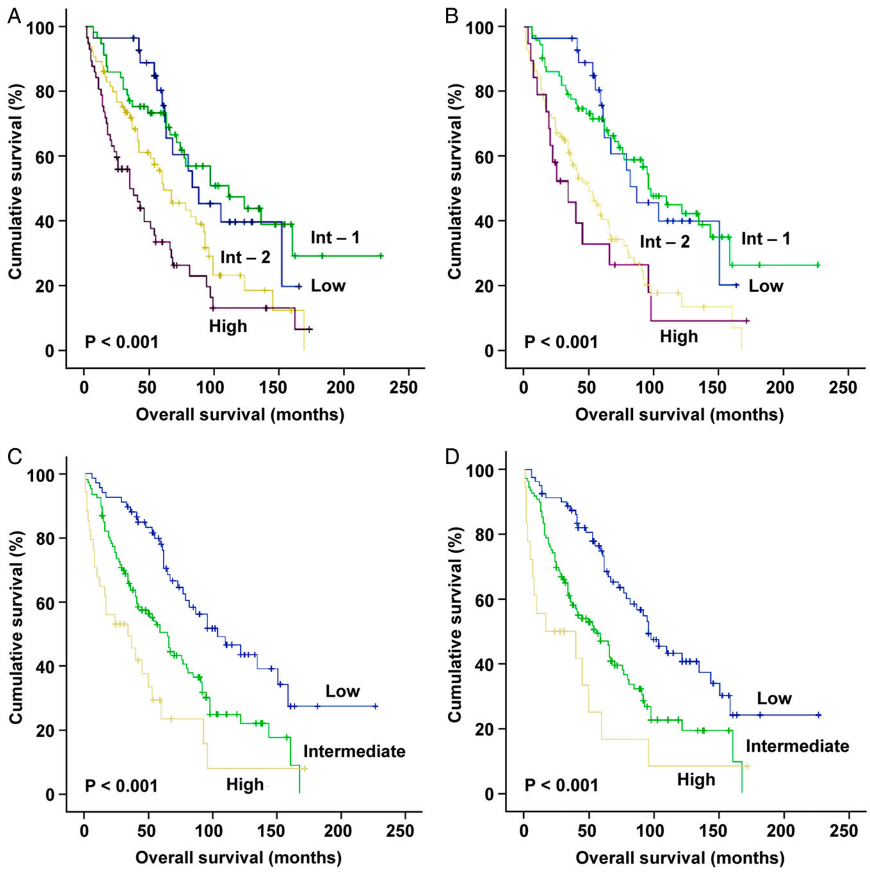

Figure 2 Prognostic scoring of 207 patients with primary MF. (A) IPSS, showing that overlap occurred between low- and intermediate-1 risk groups, and intermediate-2 and high-risk groups. (B) D-IPSS, again showing that overlap occurred between low- and intermediate-1 risk groups, and intermediate-2 and high-risk groups. (C) Modified IPSS for Chinese, showing clear separation between low-, intermediate-, and high-risk groups. (D) Modified D-IPSS for Chinese, also showing clear separation between low-, intermediate-, and high-risk groups.

aberrations other than deletion 13q or deletion 20q were independent poor prognostic indicators.[9] However, the impact of IPSS and DIPSS was not examined. In another Chinese study that examined 642 patients, more extensive analyses of prognostic indicators were performed. It was noted that Chinese patients were less likely to show splenomegaly.[7] Interestingly, splenomegaly appeared to be a favorable prognostic indicator.[7]

In this study, the OS of our cohort with PMF at 66 months was similar to that of 69 months reported by IWG-MRT.[4] On univariate analysis, the prognostic significance of age, constitutional symptoms, Hb, platelet count, IPSS, DIPSS, transfusion dependence, and unfavorable karyotypes was also validated in our cohort. Interestingly, we further identified two other unique features associated with a negative impact, palpable hepatomegaly at diagnosis, and the lack of prior cytoreduction. On multivariate analysis, age >65 years, platelet count $<100 \times 10^9$/l and blastic transformation were adverse prognostic indicators for OS. Other factors significant on univariate but not multivariate analysis might still have biologic importance, and should be investigated in larger cohort of patients. There was no difference in OS between primary and secondary MF.

Concerning leukemic transformation, there is a paucity of accurate clinical prognostic predictors. Increasing leucocytosis, marrow blast percentage $>10\%$, high-risk karyotypes, and deteriorating DIPSS during follow-up had been reported to impact negatively on leukemic transformation.[10–12] Recently, mutational profiling of a panel of genes has also identified patterns associated with an increased risk of leukemic transformation.[13] In our cohort, leukemic transformation led to death in 26 patients (20.3%), with a dismal median survival post-transformation of 4 months only. At diagnosis, Hb <10 g/dl, circulating blasts $\geq 1\%$, platelet count $<100 \times 10^9$/l, RBC transfusion dependence, palpable hepatomegaly, and unfavorable karyotypes predicted worse LFS in primary MF on univariate analysis. Unfavorable karyotypes were the only factor associated with an inferior LFS on multivariate analysis. The impact of unfavorable karyotypes on leukemic transformation has also been shown previously in Asia in a Japanese population.[9] Importantly, the IPSS and DIPSS did not predict risks of leukemic transformation. In secondary MF, circulating

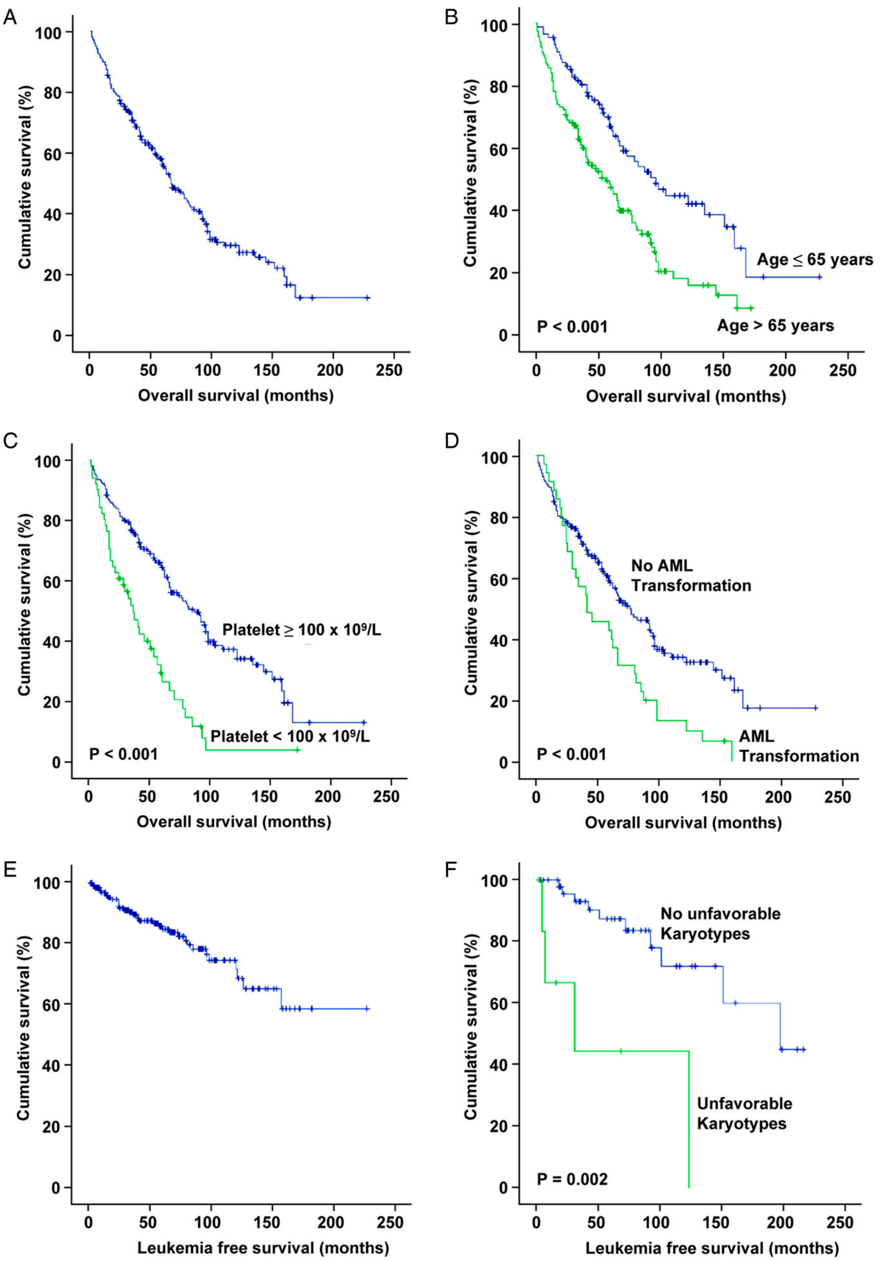

Figure 3 Prognostic indicators for survivals in 207 patients with primary MF. (A) OS. (B–D) Age, platelet count, and transformation to AML significantly impacted on OS. (E) LFS. (F) Unfavorable karyotypes significantly impacted on LFS.

blasts $\geq 1\%$ were the only factor associated with an inferior LFS on univariate analysis.

The clinical features and prognostic risk distribution in our cohort were similar to those described in major studies from Western populations (Table 2). Interestingly, a low incidence of palpable splenomegaly at 45% was described in a recent Chinese study.[7] However, splenomegaly occurred in 77% of our cases of the same ethnicity, which was comparable with those of 79–89% described previously in Western patients.[4,6] Risk distribution according to IPSS and DIPSS was also similar in our study when compared with those reported from Western patients.[4,6] However, IPSS and DIPSS did not segregate well the survivals of our patients in the low- and intermediate-1 risk categories, and the intermediate-2 and high-risk categories. On the other hand, the modified IPSS and DIPSS for Chinese segregated patients more distinctly into different risk groups. As platelet count $<100 \times 10^9$/l but not the absence of splenomegaly portended an inferior outcome, addition of a low platelet count as a prognostic indicator to the IPSS and DIPSS appeared to increase their discriminatory power for our cohorts. How the modified IPSS and DIPSS for Chinese compares with DIPSS-plus (DIPSS, unfavorable karyotypes, platelet count $<100 \times 10^9$/l, and transfusion dependence) could not be fully evaluated in our cohort,[6] because karyotyping information was available in only a fraction of our cases.

Eighteen patients underwent allogeneic HSCT in our cohorts, resulting in an OS of merely 19 months.

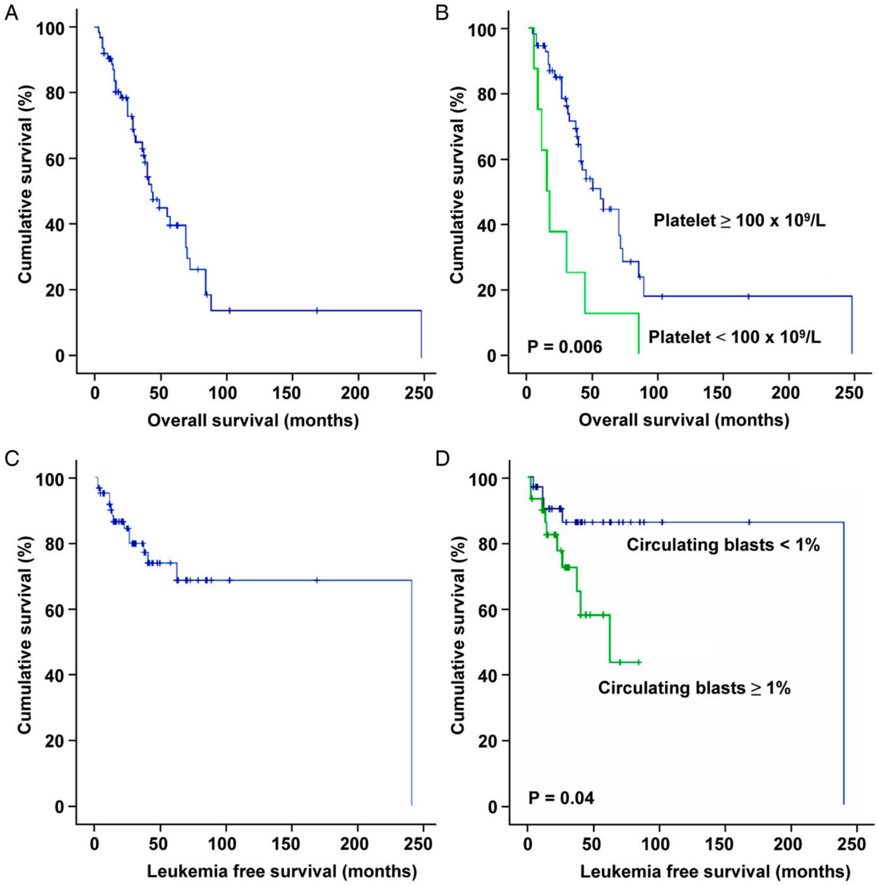

Figure 4 Prognostic indicators for survivals in 63 patients with secondary MF. (A) OS. (B) Platelet count significantly impacted on OS. (C) LFS. (D) Circulating blasts significantly impacted on LFS.

Current indications for HSCT rely on clinical prognostic models predicting poor survival or high risk of leukemic transformation.[4–6,14–16] Consequently, HSCT is often performed during advanced phases upon the acquisition of poor prognostic factors. To improve outcome, there is a need to identify prognostic factors before disease progression. Current evidence indicates that integrating the somatic mutations of *ASXL1*, *SRSF2*, *IDH1/2*, and *EZH2* into risk stratification models can independently predict worse survivals.[17,18] Mutations in calreticulin (*CALR*) have been described in patients with primary MF and have been shown to independently predict survivals.[19–21] Novel prognostic models integrating *CALR*, *JAK2* V617F, and *MPL* mutations in risk stratification at diagnosis have also been shown to accurately predict long-term survivals.[22–24] Prospective studies are needed to determine whether profiling of genetic mutations together with conventional clinicopathologic prognostic factors may best identify patients for HSCT to improve outcome.

Two pertinent issues were not addressed in this study. First, because this was a retrospective analysis, and validated questionnaires for symptomatology and quality of life (QoL) in MF have only recently become available, we were unable to quantify the burden of constitutional disturbances in our cases, and how this might be related to other clinicopathologic features. Data on symptom burden and QoL are important, as these are one of critical criteria that should be evaluated and can be improved with targeted therapy, including *JAK2* inhibitors. They should therefore be assessed in future prospective studies. Second, we have not systemically evaluated the impact of treatment on outcome. Most of our cases were treated with supportive measures. Splenectomy (primary MF: 34; post-PV/ET PMF: 19) did not impact on OS (supplementary files 2 and 6), and negatively impacted on LFS on univariate but not multivariate analysis (supplementary files 5 and 7). Since splenectomy was performed for patients with progressive splenomegaly and cytopenia, its prognostic significance might merely reflect the underlying disease status. Most patients were treated before *JAK2* inhibitors were available. Of note is that *JAK2* V617F mutation had no prognostic impact in our cohort.

In conclusion, we showed in this study that clinicopathologic features and outcome of Chinese patients with MF were similar to Western patients. Platelet count $<100 \times 10^9/l$ and unfavorable karyotypes were parameters associated with an inferior outcome,

Table 2 Clinicopathologic features and prognostic indicators of primary MF in different patient populations

Parameters	Cervantes *et al.*[4]	Gagnat *et al.*[6]	Xu *et al.*[7]	Current study
Number of patients	1054	793	642	207
Gender (%)				
Male	61	63	56	59
Female	39	37	44	41
Age >65 years (%)	45	60	18	57
Presence of constitutional symptoms (%)	27	35	21	32
Hb <10 g/dl	35	54	67	56
White cell count $>25 \times 10^9$/l (%)	10	18	15	12
Circulating blasts >1% (%)	37	60	18	26
Platelet count $<100 \times 10^9$/l (%)	17	28	46	25
Hepatomegaly (%)	50	–	17	41
Splenomegaly (%)	89	79	45	77
Abnormal karyotype (%)	29	–	29	43
Unfavorable karyotype (%)	–	15	–	13
JAK2 V617F mutation present (%)	59	66	67	56
IPSS risk (%)				
Low	22	9	16	14
Intermediate-1	29	21	43	27
Intermediate-2	28	28	31	32
High	21	43	10	27
DIPSS risk (%)				
Low	–	9	16	14
Intermediate-1	–	31	48	35
Intermediate-2	–	47	36	42
High	–	13	1	9
Modified IPSS for Chinese (%)				
Low	–	–	25	32
Intermediate	–	–	57	51
High	–	–	18	17
Modified DIPSS for Chinese (%)				
Low	–	–	28	39
Intermediate	–	–	58	53
High	–	–	14	9

IPSS, international prognostic scoring system; DIPSS, dynamic IPSS.

similar to previous observations of refinement of DIPSS into the DIPSS-plus scoring system.

Disclaimer statements

Contributors All authors participated in the study, collected and interpreted the data, wrote, and approved the manuscript for submission.

Funding None.

Conflicts of interest The authors have no acknowledgements to make and no conflicts of interests to declare.

Ethics approval Not applicable.

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